

Supplementary Material

Supplementary figure legends

Supplementary Figure S1. Clinicopathological associations. PHLDA3 expression by T category, lymph-node status (N) and AJCC stage; P values by Mann–Whitney U test.

Supplementary Figure S2. Over-expression reflects promoter hypomethylation, not copy number. (a) PHLDA3 expression by copy-number category. (b) Per-probe methylation–expression correlation across PHLDA3 HM450 probes (promoter highlighted). (c) Lead promoter CpG cg04055835 methylation versus expression and (d) tumour versus normal.

Supplementary Figure S3. Co-expression functional enrichment. GO Biological Process and KEGG over-representation of PHLDA3-co-expressed genes, and GSEA of Hallmark gene sets ranked by correlation with PHLDA3.

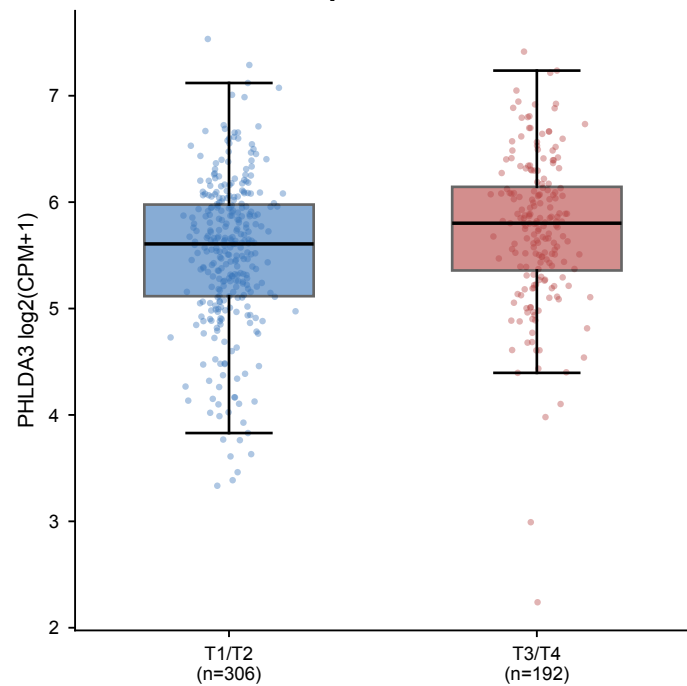
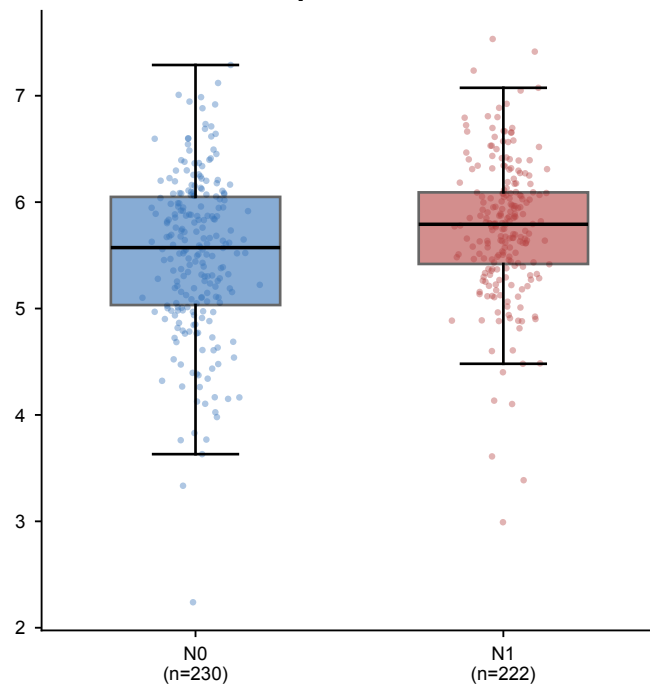
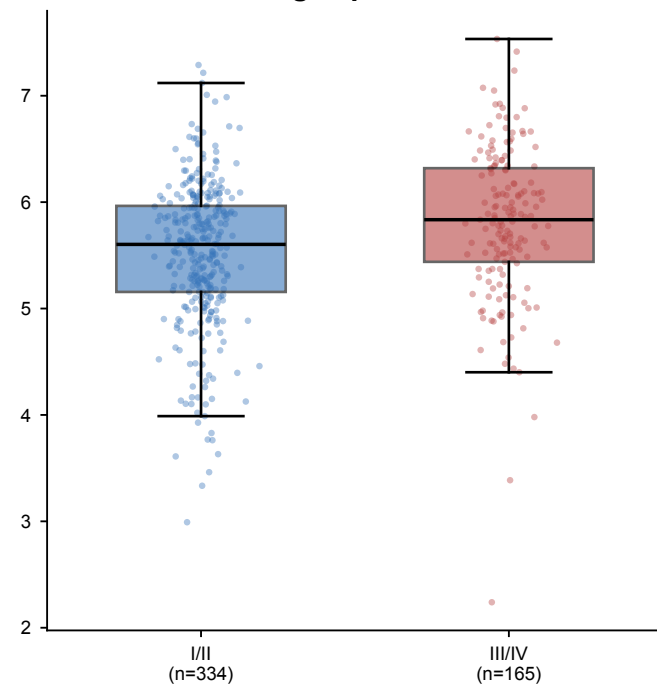
Supplementary Figure S4. Exploratory PHLDA3 miRNA/ceRNA candidate network. AGO2-CLIP-supported miRNAs targeting PHLDA3 ranked by correlation with PHLDA3 (candidate repressors highlighted), and a candidate lncRNA–miRNA–PHLDA3 ceRNA network (computational, hypothesis-generating only).

Supplementary Figure S5. Candidate drug repurposing. (a) Compounds reversing the PHLDA3-high signature (L1000CDS2), ranked by recurrence among top reversing L1000 signatures; HDAC and HSP90 inhibitors highlighted. (b) PHLDA3 expression versus drug log-IC50 across 805 GDSC2 cancer cell lines (Spearman); MEK/ERK-pathway inhibitors among the top sensitising associations are highlighted. (c) clue.io Connectivity Map mechanism-of-action-class connectivity; HDAC/HSP90 inhibitors (reverse) and MAPK/PI3K-pathway inhibitors (mimic) highlighted.

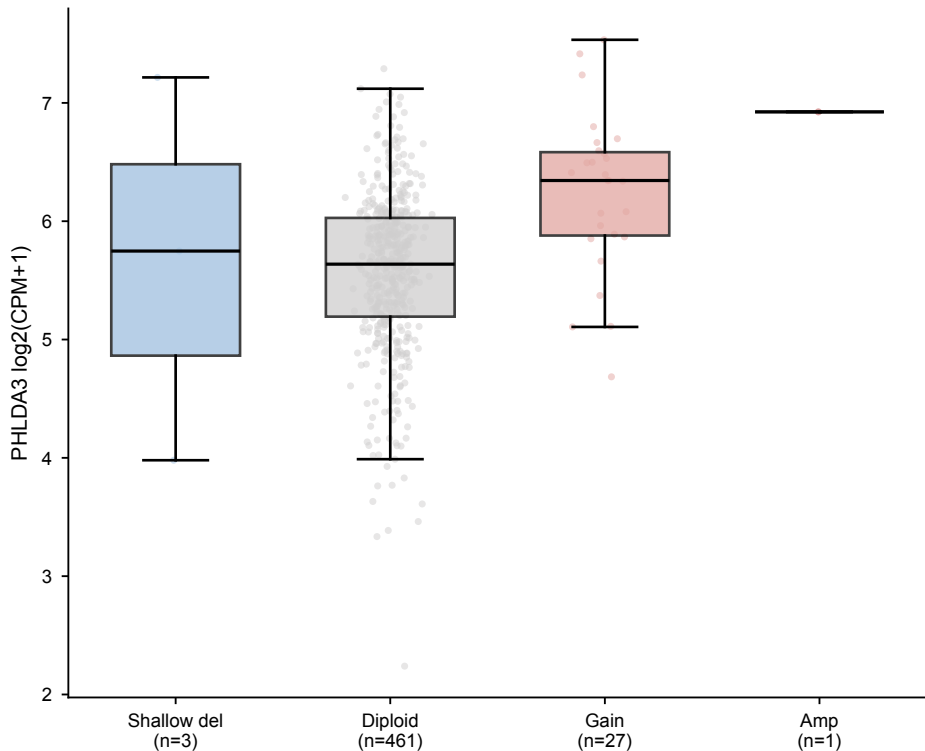
Supplementary Figure S6. Protein-level and normal-thyroid single-cell context. (a) Representative Human Protein Atlas immunohistochemistry of PHLDA3 in normal thyroid and thyroid cancer. (b) Normal-thyroid single-cell PHLDA3 detection by cell type (CELLxGENE Census), shown as qualitative context given the different annotation pipeline and single-cell drop-out.

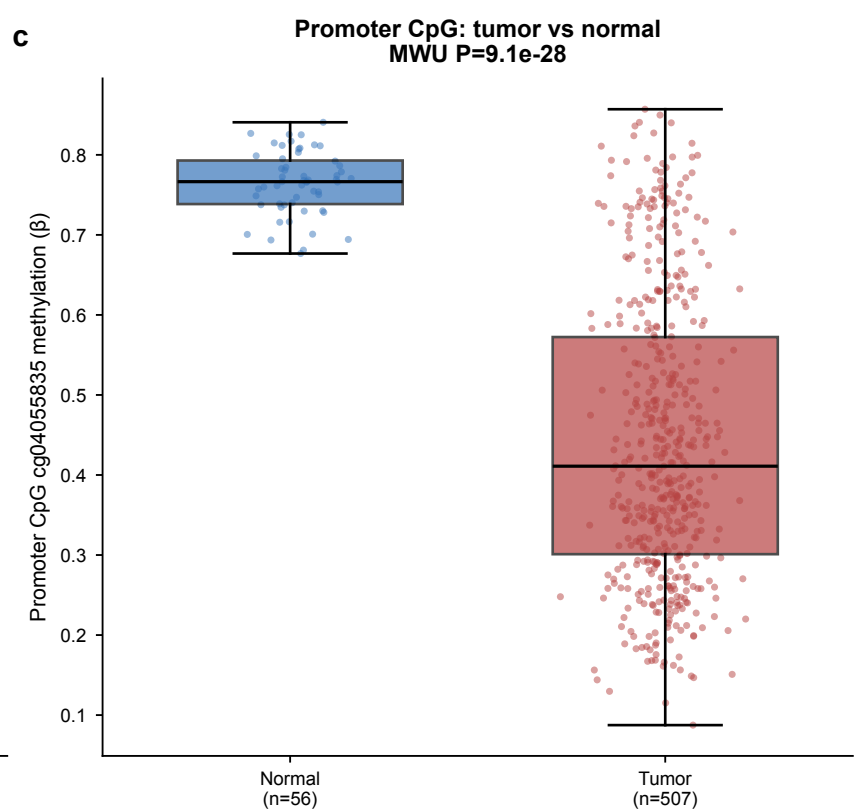
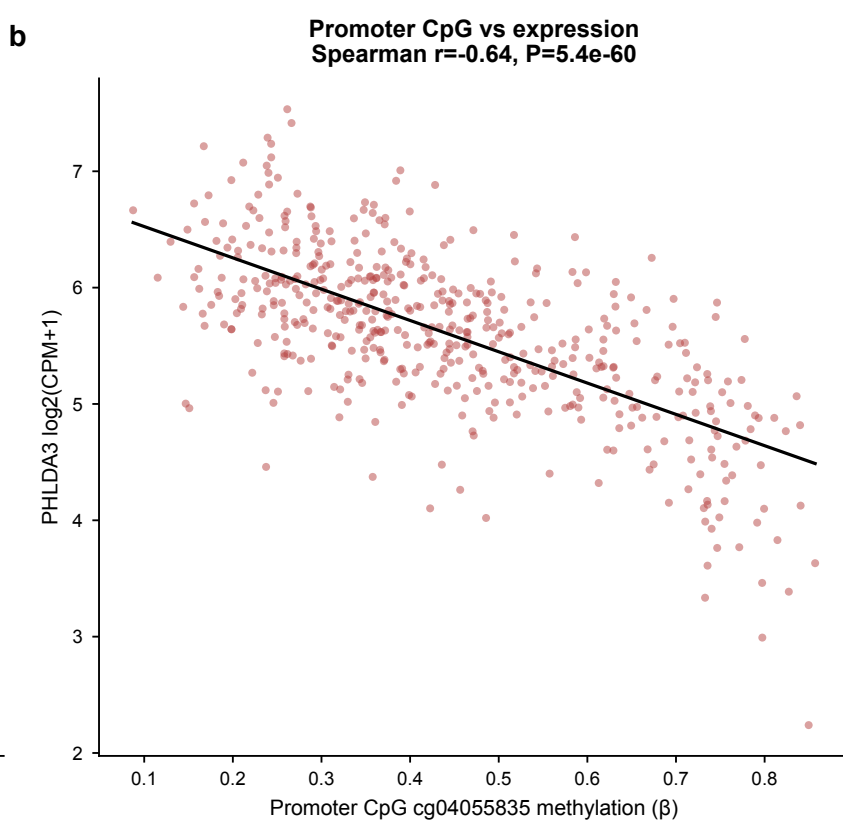
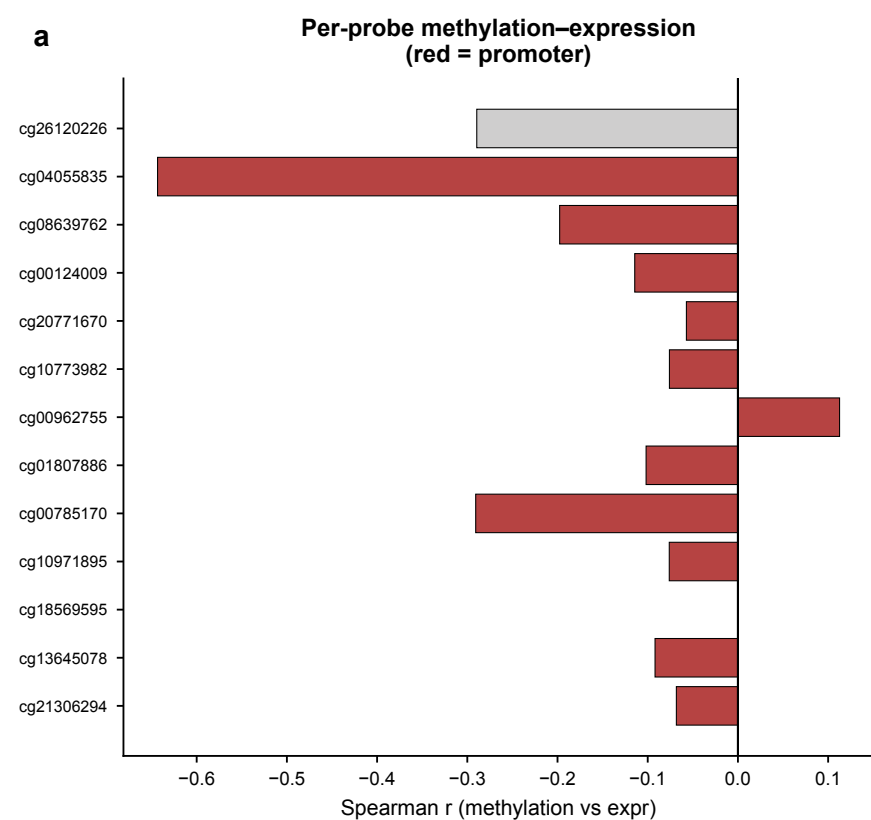
(The BRAF-stratified analysis and the sample-level single-cell comparison are shown as panels of main Figures 2 and 5, respectively.)

Supplementary Figure S7. Exploratory PHLDA3 nomogram for N1 (not a clinical tool). PHLDA3-based nomogram for lymph-node metastasis (PHLDA3 + age + T stage) with ROC (apparent and 10-fold cross-validated AUC), quantile calibration curve and decision-curve analysis. Shown only as an exploratory illustration; because PHLDA3 adds negligible discrimination over driver status (Fig. 2d), it is not proposed as a clinical risk model and was retained only to transparently show the limited clinical utility of PHLDA3 alone.

a**T** $p = 5.0e-04$ **b****N** $p = 3.8e-04$ **c****stage** $p < 1e-4$ 

PHLDA3 copy number vs expression
Spearman rho=0.21, Kruskal P=1.8e-05

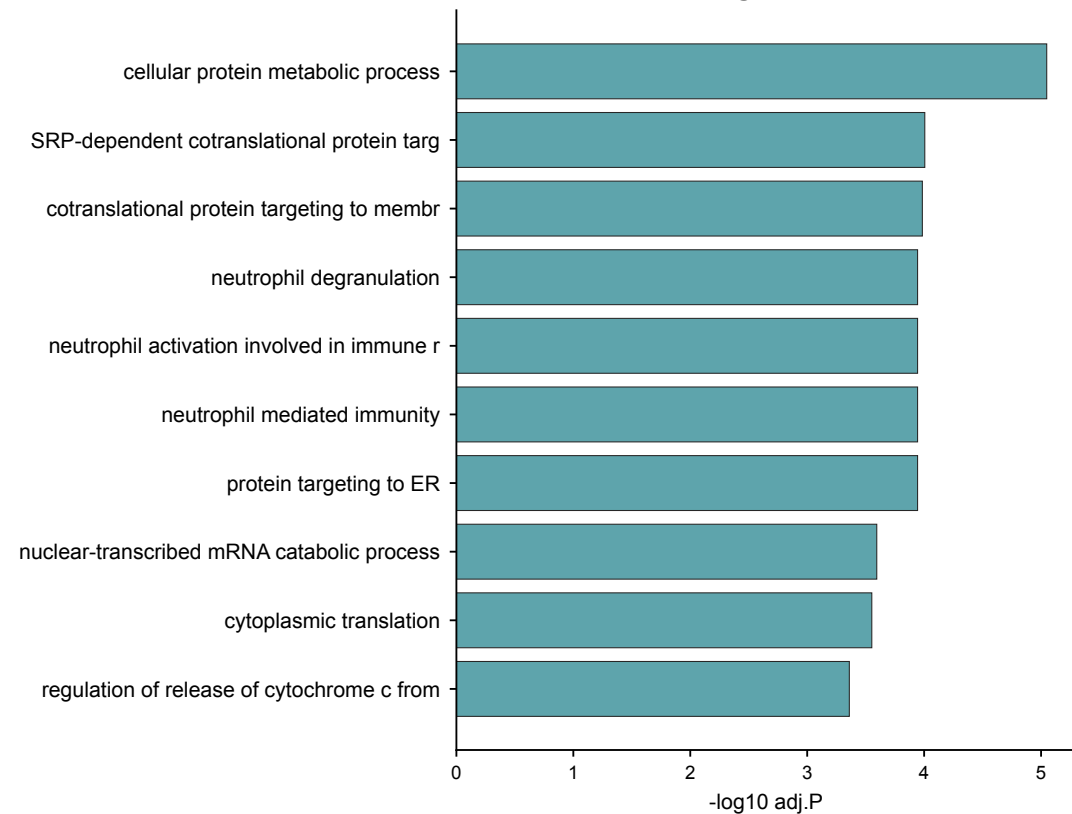




PHLDA3 co-expression functional enrichment (TCGA-THCA tumors, n=502)

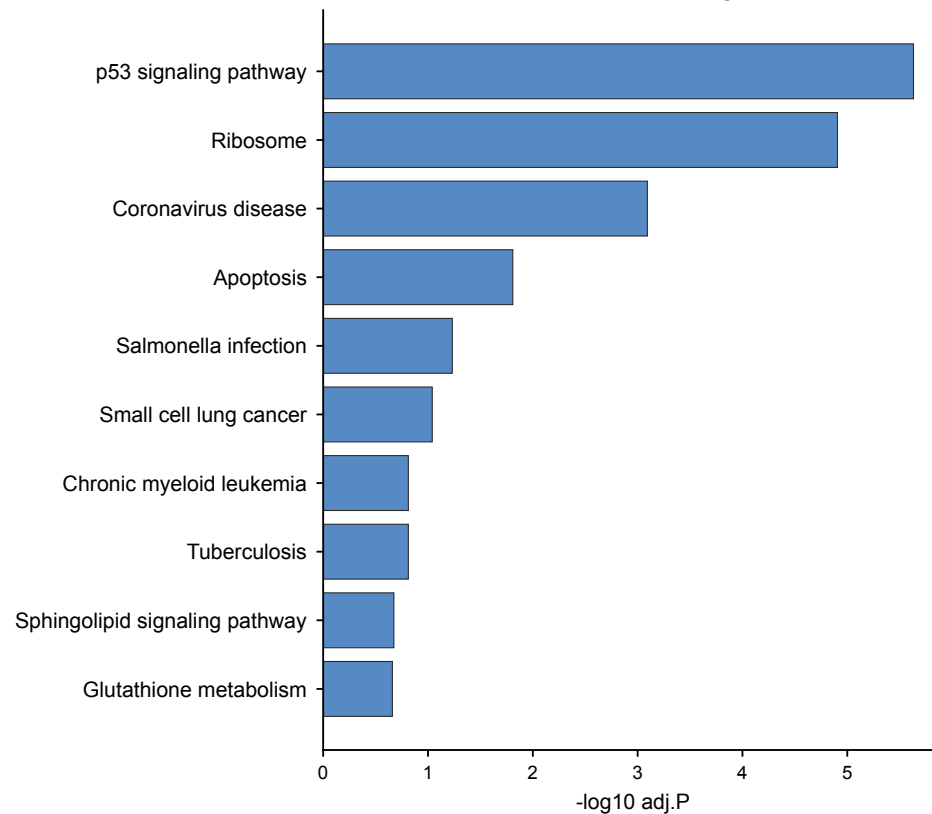
a

GO Biological Process



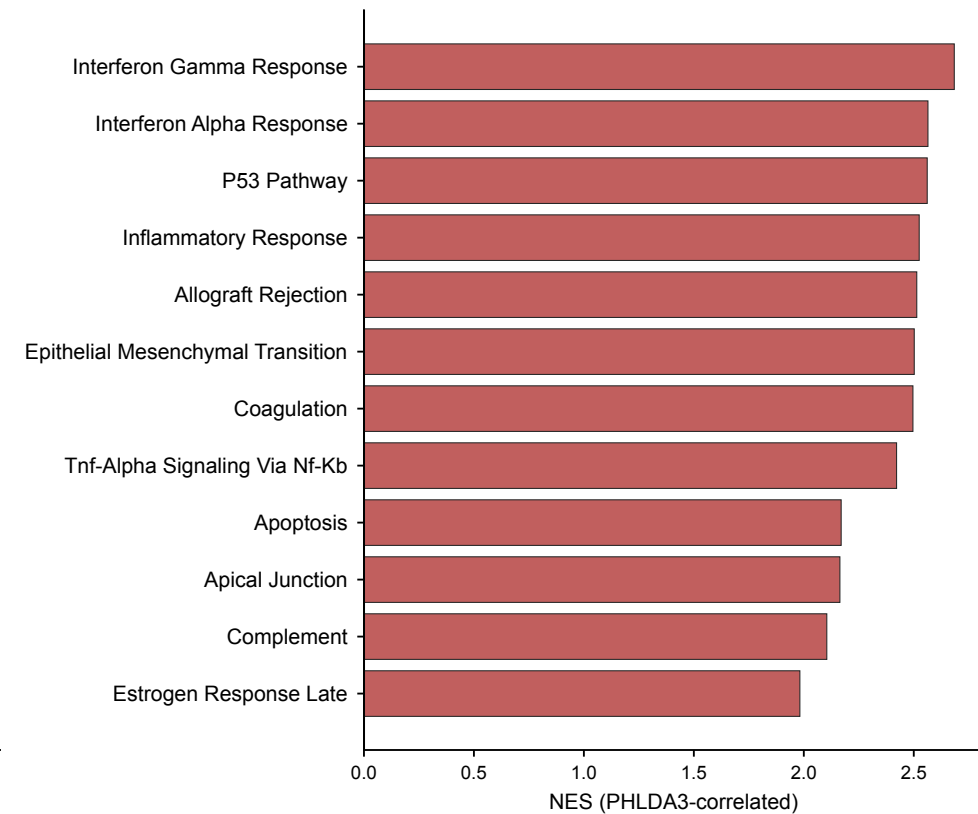
b

KEGG Pathway

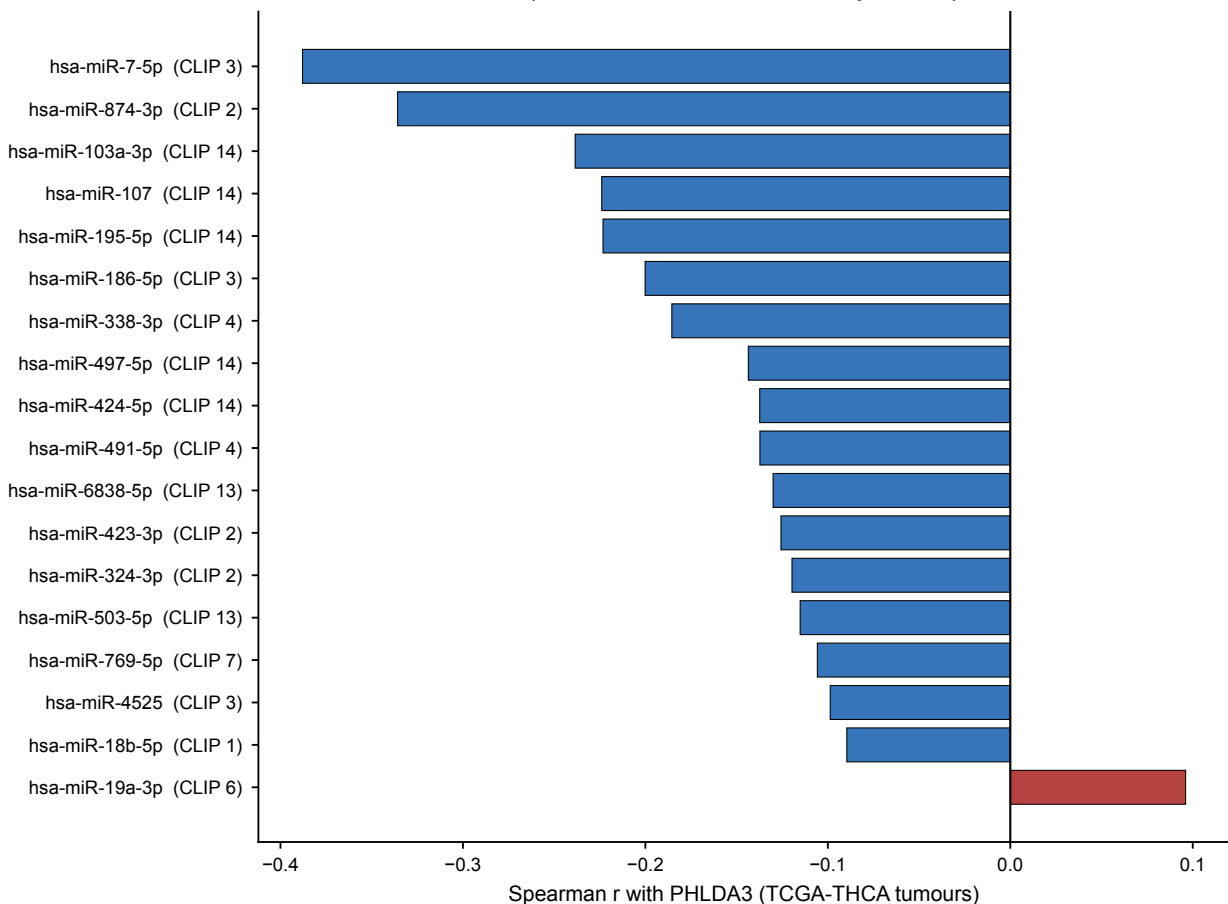


c

GSEA Hallmark



AGO2-CLIP-supported miRNAs targeting PHLDA3
(blue = inverse → candidate repressors)

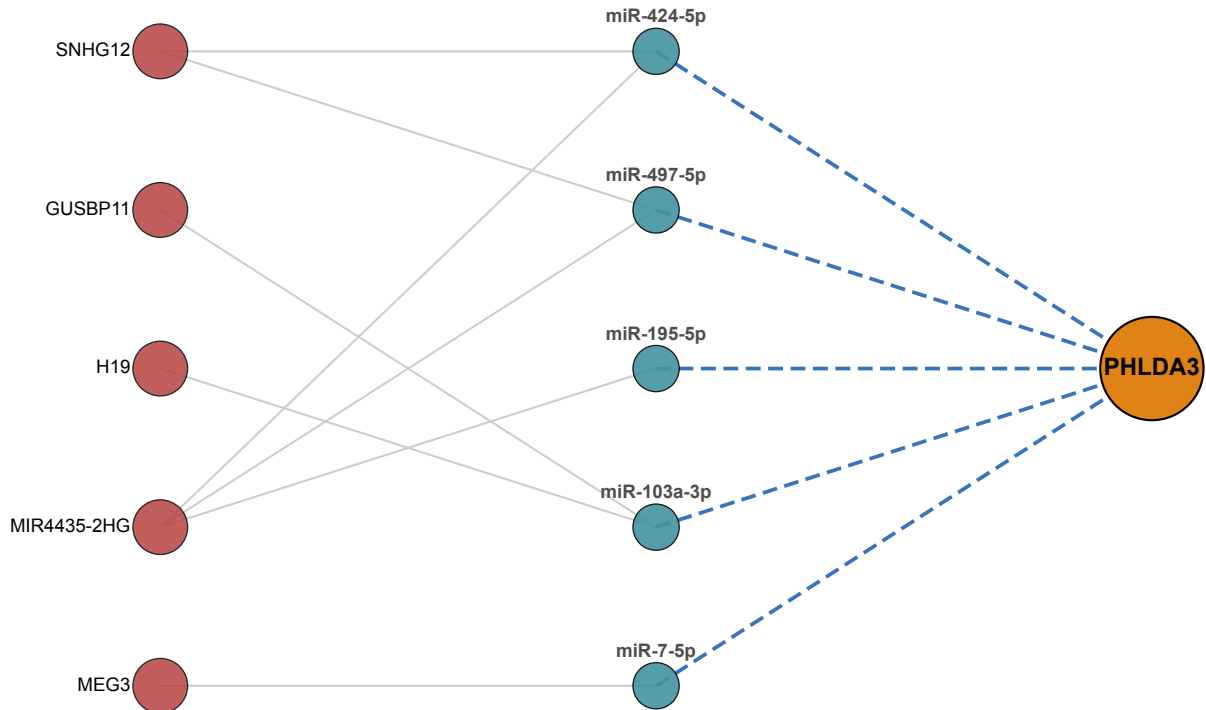


PHLDA3 candidate ceRNA network (exploratory)
lncRNA — miRNA — PHLDA3
(TCGA-THCA + ENCORI AGO2-CLIP)

lncRNA (ceRNA)
+corr with PHLDA3

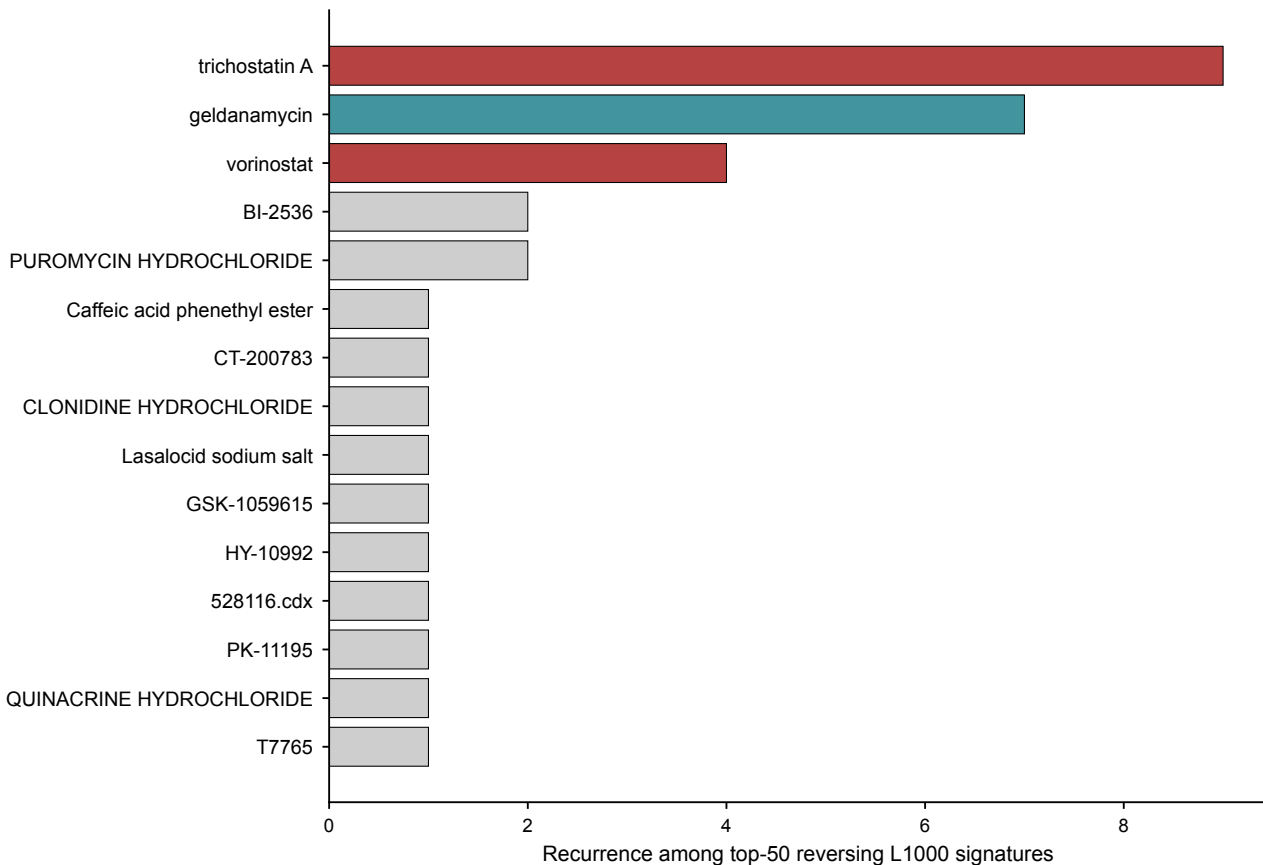
miRNA
(AGO2-CLIP, repressor)

target

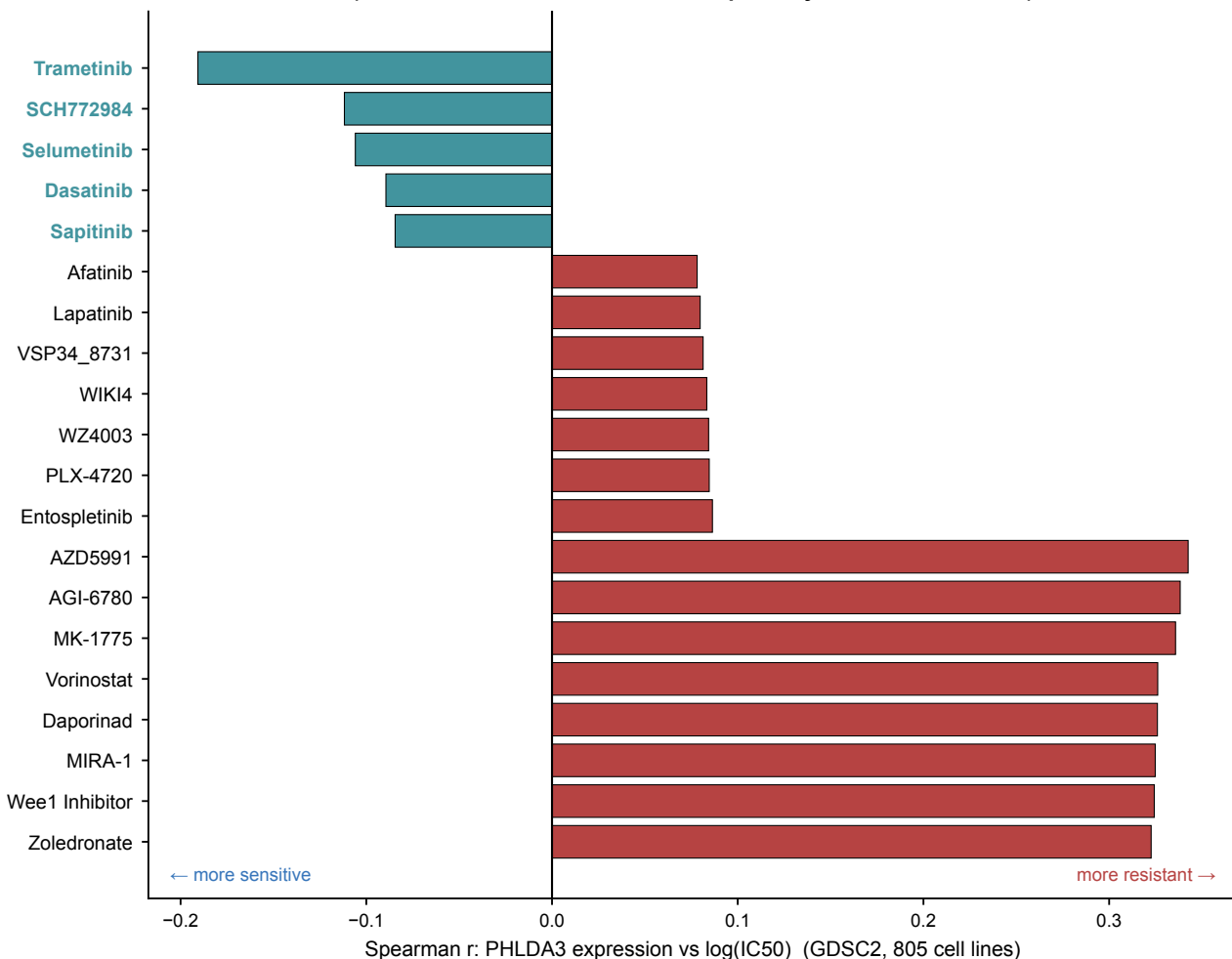


Compounds reversing the PHLDA3-high signature (L1000CDS2)

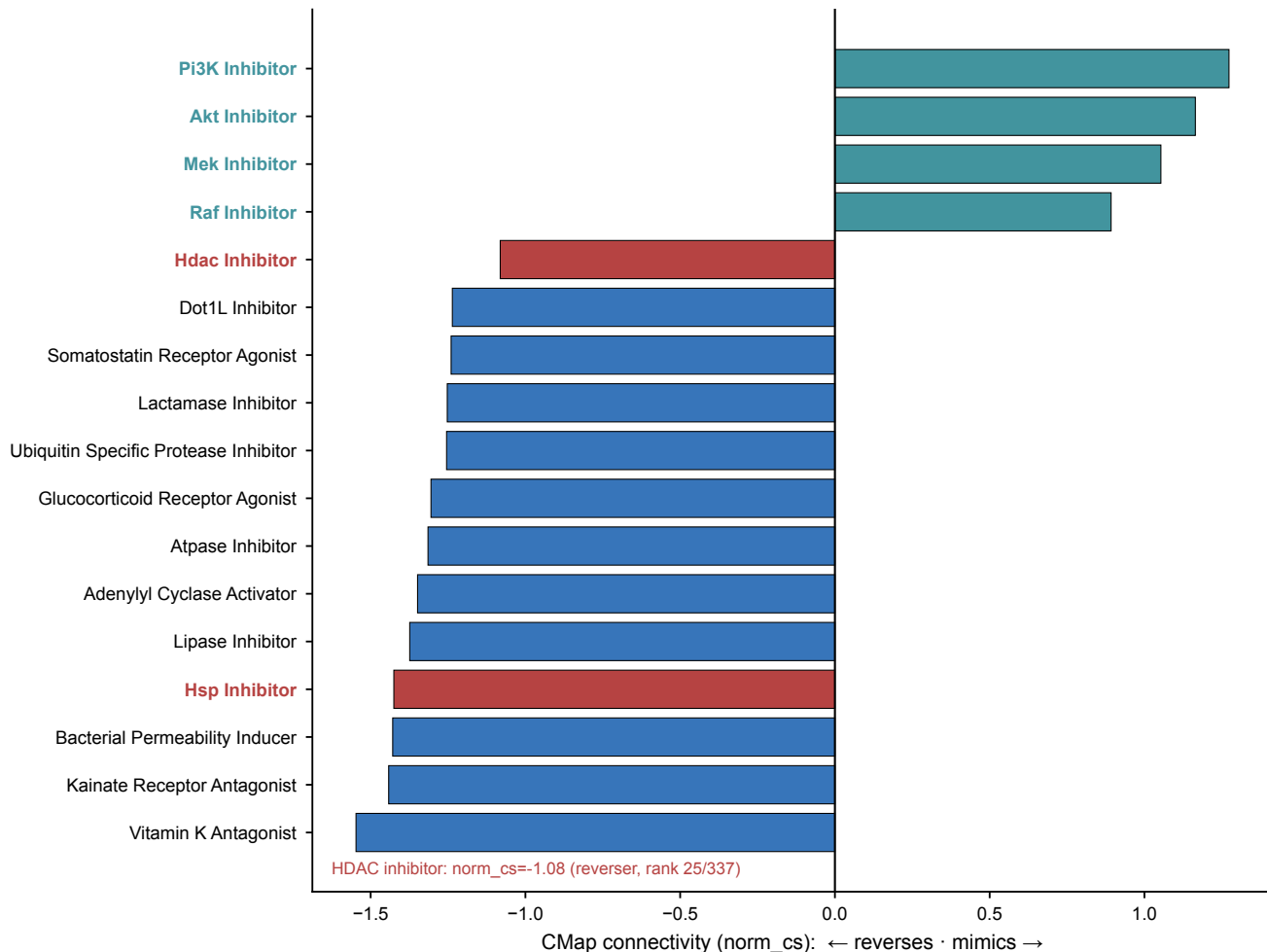
red = HDAC inhibitor · teal = HSP90 inhibitor



PHLDA3-high cell lines are sensitive to MEK/ERK inhibitors
 (left = more sensitive · teal = MAPK-pathway inhibitor; FDR<0.05)

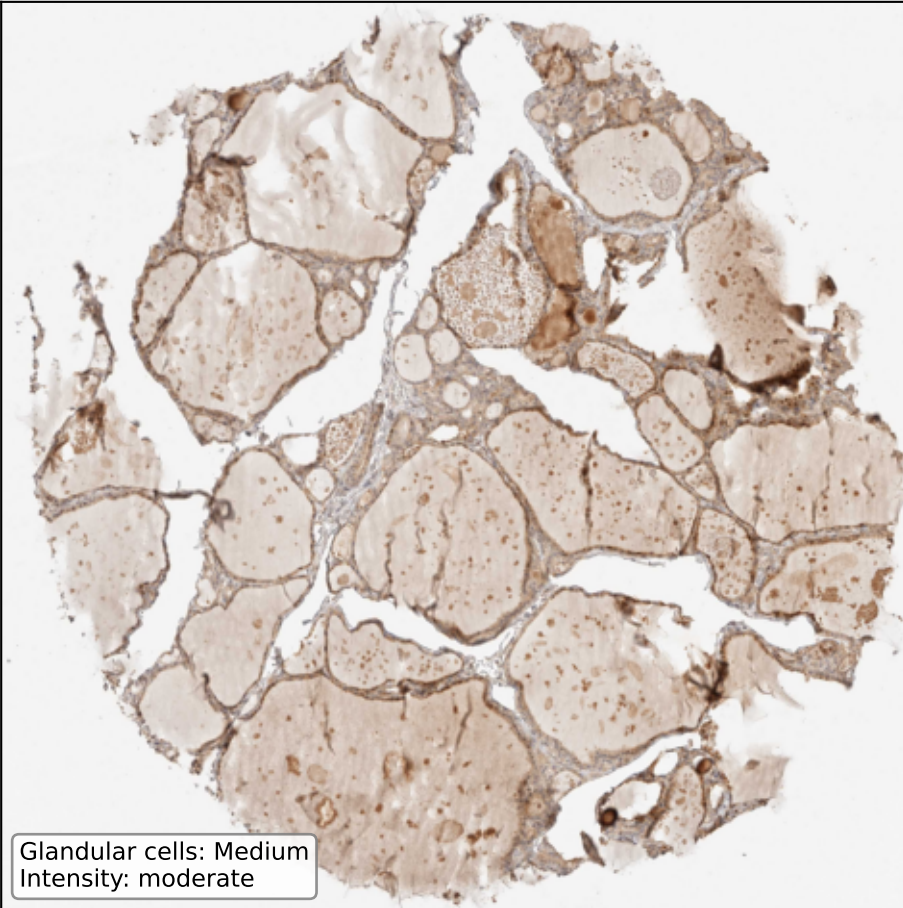


clue.io CMap (MOA-class): HDAC/HSP90 inhibitors reverse the
PHLDA3-high program (red), consistent with L1000CDS2

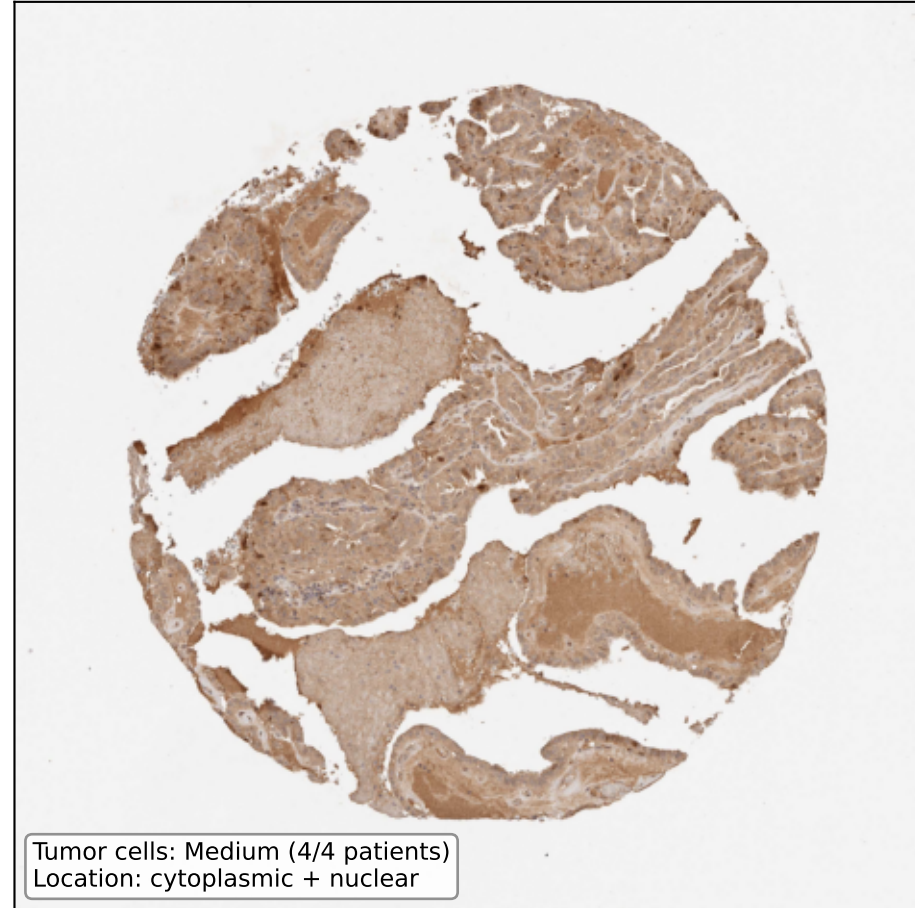


PHLDA3 protein (HPA IHC, antibody HPA017385)

Normal thyroid gland



Thyroid cancer (papillary)



a

Nomogram for predicting lymph-node metastasis (N1)

